Jensen Huang | Does AI Revenue Justify Capex, '27 AI Bottlenecks, Singularity, Open vs Closed & more

Podcast: BG2

AI Executive Summary

1. Core Thesis Jensen Huang argues that artificial intelligence represents an existential transformation requiring a global rearchitecture of computing infrastructure and human capability, driven by exponential demand for agentic systems capable of long-horizon problem solving (such as solving Navier-Stokes equations). The current trajectory predicts a massive increase in compute capacity (40–50 gigawatts annually over the next 5 years) and physical AI deployment. While supply chain constraints exist regarding land, power, and energy ("land, power, shell"), NVIDIA is positioned to lead this transition through its unique ecosystem of hardware, software, and funding, provided the industry adheres to responsible safety and regulatory frameworks like the White House Superintelligence Accord to ensure societal alignment without stifling innovation.

2. Key Arguments & Data

3. **Compute Explosion:** There is a need for 2 trillion cumulative compute units over the next five years (currently at ~1.6 billion units shipped). Next year alone will see 40–50 gigawatts of new AI compute online, which already exceeds all existing AI compute in the world today.
4. **Agent Intelligence & Cost:** Moving from regression to inference requires vastly more compute; intelligence cost is projected to fall by a factor of two or ten every year (exponential Moore's Law for intelligence), moving from ~$20 million training costs historically to hundreds of dollars. Jensen predicts long-running, agentic problems (e.g., physics modeling, drug discovery) will be solved via swarms of agents working together on common goals.
5. **Physical AI Inflection Point:** NVIDIA achieved $10 billion in physical AI revenue in the first six months of the year ($20 billion projected for the full year), representing a 100B+ market opportunity where AI interacts with the physical world via sensors, robots, and manufacturing systems.

6. **Safety Framework (Three-Legged Stool):** The industry has adopted a three-legged security framework: Frontier model review by Commerce/Department of Technology, White House Accord on Superintelligence (internal/external audits), and coordination via the SI Force. Safety is treated as orthogonal capability that accelerates equally with functionality ("functionality + safety = one thing").
7. **Financial Strategy & Market Consensus:** NVIDIA trades at 13–14x earnings despite a forecast shift from flat/shrink to 74% growth in 2027; the market consensus was off by 200% last year. The company plans to spend over 50% of free cash flow on buybacks (60%+ in one quarter) while building long-term value via software licensing and ref shares.

8. **Fungibility & Infrastructure:** NVIDIA aims to transition from a GPU vendor to an infrastructure company where "fungibility" in workload, models, and customers is the primary competitive moat. Software licensing and ref shares (refunding share) are emerging revenue streams independent of hardware sales, leveraging millions of installed GPUs and enterprise SLAs.

9. Actionable Takeaways

10. **Investors:** View NVIDIA not just as a hardware provider but as an ecosystem and infrastructure play. Focus on long-term CapEx build-outs (~$4 trillion annually by 2029 for knowledge work) rather than short-term volatility, recognizing the shift towards software licensing and ecosystem fees alongside hardware sales.
11. **Industry Leaders:** Integrate safety and security technology directly into product development rather than treating it as a post-market add-on. Adopt the "zero-trust" containment model (similar to parenting or hiring new employees) for AI deployment.
12. **Policy & Regulation:** Understand the U.S. "Three-Legged Stool" governance structure as an actionable, tangible framework that balances rapid advancement with safety oversight. Engage constructively in these discussions rather than waiting for regulations to be written from scratch.

13. **Business Strategy:** Companies aiming to deploy AI agents should utilize NVIDIA's Omniverse digital twin of the physical world to train and test autonomy before deployment, ensuring durability and "low risk" fungibility that allows enterprises to scale without single points of failure.

Full Transcript

This is an existential issue for us. We could easily stop, but if we stop today, we lose. Functionality, capability, safety, security, they're all at exactly the same place. Priority number one. It's our third pod back together. In the last 10 days, you've been at the She State Dinner at the White House. You were at the White House summit where we did the accord on superintelligence. And just yesterday, you were there with your parents when the president... presented you with the National Medal of Science. You were the only investor at the table. You were the only investor at the table. Now that we reflect it on it. Well, tell me, being there with your parents yesterday, it struck me five out of the six people on stage were all born outside of the United States. How did that feel? Well, first of all, It's a great honor all by itself. And it's a reflection of the contribution that the company has made. And obviously, we invented a new computing model. This is probably the most pervasive computing model in the world today. It's the only computing architecture that's in every cloud. It's every computer company. It's used in every field of science. It's used by gamers. It's used by creators and builders. And it's used by scientists. It powers intelligence.

I mean, just if you take a step step back and you look at what the company has accomplished, you know, we started out building a graphic ship for gamers and I'm still proud of that. And of course, and we now shipped one and a half 1.67 billion billion units. Right. That's how many generations of I mean, I raised generations of scientists and kids and designers, you know. And so, so I think the impact that and here we are, you know, we're the infrastructure that powers the intelligence of society. And so, so it's a, it's an incredibly proud moment for me to receive the award on behalf of the company. Now, for my parents, you know, I wanted them to go in and I gave them literally, you know, 24 hours at the time, and they don't travel that much. They don't travel that much. And my dad didn't even have a suit. Wow. Yeah. So he ran out and bought a suit. He looked great. They looked great. Yeah. It was a brand new suit. And he showed up and I didn't realize he was He had a suit. I thought he was just gonna wear some shirt. And so he went out and bought a suit. My mom got a new dress, you know, and literally in 24 hours. Oh, I know how it works. Oh my gosh, it's incredible.

And so we got him on a, we chartered a plane for them, got him there. And they were so happy to meet the president.

President Trump walked my mom because she has a hard time walking and the two of them were were walking off stage and he held her and Walked it walked her down the stage and they were just incredibly incredibly proud to meet her. You know and But for them it's their American dream, you know my dad I still remember I think I was like five years old or something like that and and he had just come back from the United States on a business trip. He was working for an air conditioning company. You were living in Taiwan? We had just moved to Thailand. We were just about to move to Thailand. We were in Taiwan. He had come back. He was in the United States working for a company called Carrier, their air conditioning company. He was here for training. Of all the places in the world, they landed him in New York. You can just imagine.

a kid in the late 20s, I guess late 20s, and never been outside of Taiwan, goes to United States, New York City of all places. And he saw the future of the future. And he came back, you know, mind blown, all the things that he bought at the airport and brought it back. He brought chocolate, he brought... You know that cheese that you have in a tube back in the old days is kind of strange. It comes in a tube. That's great. And he brought back all kinds of strange things. And he says, we should go there someday. And the first opportunity they had, they sent my older brother and I to United States. And it was his dream. I don't think many people realize that you came to the United States ahead of your parents.

They joined a few years later, but it just hit me like a ton of bricks that five out of the six people on that stage yesterday were born outside the United States. You and I have talked about the need for, you know, with all the other stuff going on, a real strategy and an immigration policy policy that makes certain that we're still the place that all the great builders in the world want to come build. This is America's superpowers. Right. In a lot of ways, the intellectual property creation capability of our nation is unlike anything else in the world. And the reason for that is we attract the world's best. We are the place where they want to come and build their lives because of the opportunity it provides, the freedom that she offers, the opportunity that it

provides, the spirit of invention and entrepreneurialism and the can-do attitude. In the United States, everything is possible. We start with, of course it's possible. Of course we could do that.

Instead of, you know, I do a lot of travel and of course you can't do that. Of course, you know, in other places and we take it for granted. But the spirit, the energy that comes around with the culture that we've created here, it just can't be understated. And so I think the flow, the inflow of intellectual property into this country is unlike anything else. the world's ever seen. And we want that. We want all of the world's great minds to come here, bring their great ideas, make them, realize them here, realize them here, and let us keep that flywheel going. It's a delicate system. It's a delicate balance I get. It's a delicate balance I get. And so we have to do it lawfully. We have to do it rightfully. But we surely want to do it.

Yeah, for sure. So I want to start high level, just kind of where we are. Then I want to dig down a little bit into the safety and regulation and getting this right, where we are in terms of the compute build out. And then finally, we're going to finish up drilling in a little bit to NVIDIA's growth and what's going on there. But you've been really good at calling the future. Two years ago, we were sitting here and you said that the market was dramatically underestimating the amount of compute build out that was going to be required and was going to happen over the course of the next five years. You said it's not going to be a trillion, it's going to be two trillion at a minimum in terms of cumulative compute build out. And of course, now we're over that. You said before everybody was thinking about it that inference, compute was going to 1 billion X because of inference time reasoning and thinking. And then last year, of course, Before we saw Opus 4.6, before we saw Cloud Code, you said we're entering the age of agents, right, where we're going to unlock a lot of these capabilities. And of course, we saw Opus 4.6 and Cloud Code really took off. So where are we now, right? We have Elon and Sam and Dan. There are only three things that I said that came true. There's another 200. Why don't you just keep on going? Well, I just let that be the podcast. So our investor day in a couple of weeks is on, you know, entering the singularity, this acceleration. We have Elon and Sam and Demis all say, hey, we're in the foothills or entering the singularity. We have, you know, model-solving Navier stokes, as you know.

So when you look into your crystal ball, where are we and are we inflecting? Are we accelerating, you know, into a different phase of superintelligence? And so let's unpack that. First of all, It's very fairly logical why the amount of compute that we're gonna need is a billion times more than where we were just recently the sick caller last year and the reason for that is because Thinking and problem-solving takes a lot more you know inferences then Repeating something regurgitating something that you memorized. So when you're a chat GBT you give it a prompt, most of that knowledge was encoded into the large language model. Unfortunately, of course, it hallucinated. And so the next click was, it was a generative, but it was also grounded or conditioned by ground truth. And so it would have to go out and do a search, do a quick search, comes back with the ground truth. And then from that conditioning, augment it with his own what it already learned pre-training, and it would generate whatever the answer is. Well, most of the things that we want to really do that are interesting requires you to reason step by step, to come up with several options, plan, and you might decide, let's see how these things all roll out. And then based on the answers of all of those, I reflect on the best answer so far, the best path so far, I might choose one or two or three of them, roll them out.

Okay, prune the rest of it and then just keep on doing this and you know people we do this that's the reason why why we say great strategists love strategic optionality You don't want to prune things too fast on the one hand, but you you also have to optimize your resources Well, AI's are doing that. Okay, so so they're keeping their strategic optionality. They're rolling things out. They're pruning They're they're reflecting their pruning so and so forth and all of that thinking All of that exploration just takes a lot more compute. And not on top of that, because it's able to now do productive things, we're gonna use it for more things. And so the compounding of both, it uses more compute to do its job, but it's doing a good job, and therefore we're gonna hire it to do more jobs. And so that all compounds. At this point, I would say that the big observation from last year is that, On the one hand, we want the models to get better and better, and we still want the models to get better. The scaling laws of models are going to continue. However, the attachment of a harness, basically the exoskeleton of the large language model, kind of the body of the brain, that allows it to manage

working memory, manage knowledge. How do you go from working memory, the short term, the long term, to basically...

formulating your knowledge graph, which is what we all do. We kind of think about, hey, you know, what did I learn? And how does it fit into everything that I know in the past? And, you know, how we go to bed, we do compaction, things that don't make sense. We kind of don't matter. We kind of eventually forget, but the things that kind of matter, somehow it gets reconnected. And so we're doing that kind of offline. And so we got working memory. We have offline knowledge extraction and creation, knowledge graph creation. that whole part of memory is a big deal. The other thing is, of course, tool use. And we've taught these tools, these AIs, how to interact with the most important tools that humans have ever made. It's no different than if you're a robot, it comes into our house, it's more likely you're going to use all the tools that I have in the house than to go off and...

build it from scratch. And so, so Spice is still going to be useful and Verilog will be useful and simulators will be useful and, you know, design compilers and, you know, Photoshop and AutoCAD and, you know, the cell Katia and all the Siemens tools, all of these different tools that we've been using to run our company, build our companies, design things, operate things, they're all going to be great tools for the agents to use. But the agents had to learn how to use it. Right. And so these two areas knowledge knowledge and working memory and tool use huge unlocks in the last year. So now all of a sudden you could you could put these large language models encapsulated in their harness or exoskeleton and you can put them into your company connect them to the information, the ground truth that you have, connect them to the tools that they can use, and all of a sudden they start doing amazing things. Now the easiest part, frankly, was coding. That's kind of the first on-ramp to useful work. And in a lot of ways, coding is easy because both the customer and the tools that they use are fairly accustomed to things that has tremendous use, but are not extremely easy to use. And so in the early days of coding, it wasn't extremely easy to deploy, but software engineers found a way. And so now I think the next click for us is now, and you guys have probably been playing with Astra, OpenAI, so Astra is really fantastic and we're using it in all these different places. And

so now the next phase of these agents, they're able to do you know, design work and a lot of creative marketing work and we're using it to go, you take a picture and give it to Astra. It literally recreates it into CAD. It recreates it into the CAD, which is incredible because we need the CAD so that we can interact with the CAD and make fine tuning and adjust. And so Astra could really give us the first initial implementation. Incredibly productive.

And you take Astra and the latest release and you apply to all of these different domains. I think that besides software engineering, we're going to see designers and builders and creators and this whole area, scientists and all of this whole unlocking. Now, the challenge, of course, is how do we deploy agents safely and securely into the world's enterprise? This is exactly the same question again with safety. security and we'll talk about it. What I'm struck by is that all of this occurred this year with OpenAI and Anthropic having, I don't know, two, three gigawatts of compute each. So relatively small amount of compute relative to what's coming. And so again, what I'm interested in is the way the economy generally works or people model the economy or they model a company is linear deceleration. They're not very good at modeling acceleration. But here we have this double exponential. We have massive amount of new compute coming on board. So if OpenAI and Anthropic are, I don't know, three to five x-ing their compute next year, does it stand to reason? I mean, we solved Navier-Stokes.

We'll look back at this and say that was a tiny amount of compute and intelligence, the worst intelligence we're ever going to have from here forward solving a millennium problem. So when you look at what's coming online next year, do you expect that we're going to have major breakthroughs and long running, agentic problems that we can use these swarms of agents on? Well, Navier Stokes is really a great example. Remember, OpenAI did not solve Navier Stokes from nothing. Correct. They solved it from the foundation of a lot of great scientists and a lot of great mathematicians and a lot of great previous work. Earlier I said we're at kind of the tipping point of a lot of these different fields because we're kind of at that phase, that point with Navier Stokes with a great deal of foundational work already been done. If we could just apply the exhaustive exploration of AI to the previous done work, we're gonna see one

breakthrough after another breakthrough rather quickly, but we do need a lot of compute to do that. And the reason for that is because a large amount of exploration was necessary. If it was simply somebody resolving some fundamental first principle theory in order to discover the breakthrough, if it didn't require a vast amount of exploration of space, search space, not space space, but search space. If it didn't require that, then somebody would have found it earlier. You know, people are pretty smart. And so I think that there are many problems like this that are just on a precipice and really wants to just have that massive amount of intelligence exploration to get us there. In the coming years, you know, my expectation is that the cost of intelligence would fall fairly quickly. And the reason for that is because obviously we're bringing more compute online. These models are getting smarter. When you're smarter, you could be more efficient. And so the combination of these two things, the cost of intelligence will come down, which is, of course, what they want. And this is no different than Moore's Law. If you go back 30-some-odd years, we have three decades of computing cost decreasing by a factor of two every single year. And we want intelligence cost to decrease by a factor of two or 20 every single year because the consumption of it is going to more than make up. That's right. More than make up for it. And so there's a massive empty space in the world that wants to be filled, empty vacuum of opportunity. And we just got to get the cost down.

I guess what stands out to me about Navios Stokes is that they had 10,000 agents working together. To me, that sounds like a hyperscale company basically all pointing agents to work together towards a goal. And I think in the early days right now, it's a lot of paralyzed search exploration. But I think the goal is that more of these Agents can work together to a common goal and that we can solve a lot more and to your point You know, I think our math says probably cost about four million dollars of compute to solve solve, you know this instantiation of Navi Stokes The cost will the cost to do something like that over the next few years is going to drop by another 5x 10x right so the number of things that we're going to be able to do is Is rapidly our ambition is going to go sky high The original GPT probably cost let's pick a number say You know maybe 20 million dollars to do if you wanted to train GPT 1 now You'll probably cost you what a thousand a hundred dollars. Maybe it's like a hundred bucks Okay, and so you you

went from you went from 20 million dollars. Let's say 10 million dollars to a hundred bucks and we did that in 10 years. Yeah, okay, and so I fully expect, meanwhile, our ambition got so large that the models that we're training now, 3 trillion, 10 trillion, 30 trillion, we're talking about gigantic models now. And we want it to be, and so our ambition for the capabilities of technology has gone up. Meanwhile, our use of it has gone up between 10 years ago to now, probably, I don't know, a million times. And so I don't think this trend is gonna stop. I think there's a, there's a fairly insatiable demand for intelligence. And the smarter you are, the more intelligent you are, the more ideas of applying intelligence you have. The more I've learned in life, the more I realize there's more to do. And that's because every time you learn something, it opens a new door. Every time you learn something, it opens a new door. And every time you open a new door, you see more opportunity, you see more challenges, you see more problems.

And so that's the future. People say, we're going to have more time. I completely disagree. I think for most people, this intelligence is going to unlock more opportunities. And they're going to want to pursue it because they're ambitious. Give me an example of a family, a father, a son, a husband, a wife, a mother. Who don't have ambition? for their families to have more. It's just, they just don't have the opportunity and these things are going to open up opportunities. If you go back to 1930, Keynes talked, he made a prediction about the industrial revolution. And he said, you know, in a hundred years, the economy will have 8X, but the work week will go to 15 hours a week. He was right. The economy did 8X, right? We had massive abundance and productivity gains from it. Yeah, except we're working eight hours a week. But we didn't go to a 15 hour week. And what he got wrong was the human spirit.

Right, he didn't realize that that work is purpose. Yeah, discovery is purpose, right? It's deeply encoded in I think our DNA and I'm working harder than I ever have having more fun than I ever have not because I have to like my grandparents did, right? I'm doing it because I want to. And so I agree with you on that. I'm fascinated with this idea. The singularity postulates that we kind of now have all these agents that can work together on long running tasks. They can stay on tasks. I love your keep each other on task, working as a team. Hey, you're off. That's right. I love your idea

that, you know, we build on top of all the things that have come, right? And so if I think about full self-driving as an example, we talked about it for a decade and we really didn't get there, right? But all of a sudden, like you go on Twitter, everybody experiencing Tesla FSD, experiencing Waymo, We're here. Somebody asked me how was the drive here this morning. I said, I don't know because I got in my car and I hit FSD and I didn't pay any attention until we pulled into the parking garage. I was working the whole way here. But if I think about that in the fields of science, the breakthroughs that are going to come, in the fields of pharma and genetics, we've been laying down the tracks for 30 years. We're going to break through on a lot of massive cures for the world, of course.

ChatGPT is already helping parents plan their day and solve their kids' health issue and helping them get better educated. Talk to us a little bit, though, as to why that's having such a hard time breaking through, Jensen. We have all of these things that are out there improving lives and that are right around the corner. And so that's kind of curious to me as to why that's not breaking through. I guess let me take a shot at it. And, you know, we'll ultimately not really know. I've spoken to a lot of people. They told me what they thought. I mean, one of the things, of course, is AI or super intelligence is soft matter. You can't see it.

You've got the whole world talking about this technology, but you can't see it. You can't touch it. And even though they're already benefiting from it, when they do search, they're already benefiting from it. When they go shopping, they're benefiting from it. When they're using Netflix, they're benefiting from it. You go to Walmart, you're benefiting from it. The fact that FedEx is getting stuff to us. It's all leveraging super intelligence and accelerated compute today. But we don't do a very good job telling that story. So life is, they're benefiting from it. It's invisible. Right. So if we shut off the data centers and shut off super intelligence for a weekend, I think everybody would notice. Right. I know you had a question about robotics in particular. Yeah.

I guess you've notably said a few times this year that the chat GPT moment for robotics is here. And I think you've also said that it could be your next 100 billion market. So I guess what are you seeing now in the inflection

point in this year that this is around the corner? Yeah, I understated the 100 billion because we're already the first six months of this year, we did 10 billion in physical AI. Physical AI is basically AI that's somehow interacting with the physical world and it could be photon counting CT machines. It could be ultrasounds. It could be new imaging radars. Next generation of 5G. Not only does it understand radio waves, it's going to understand signals of all kinds. It's a sensing network. Of course, robotic self-driving cars, manufacturing systems, visual inspections.

geospatial satellites imaging the world and reconstructing it into 3D. And so basically, if you look at the entire electromagnetic spectrum, there's AI stuff going on from visible to invisible spectrums. And we're doing work across that entire thing. And as an AI, there's the perception part of it, there's the reasoning part of it, and then there's the planning and actuation part of it. planning and actuation, then that's self-driving cars and robotics and things like that. In warehouse delivery vehicles, pick and place robots that you don't have to program as precisely because it learns by itself, all the way to people talking about consumer robots coming into your house, making your bed and things. Or healthcare robots, hospital robots, lots of really good examples there.

precisely delivering exactly your prescription to the patient, having robotic care, literally 24 seven every single second to keep patients safe. And so just lots and lots of great examples, entire surgical rooms that are going to be completely robotic. And so lots of great technology being deployed. And so if you look at this whole thing, we're the world's largest physical AI company. We did 10 billion in the first six months. We did 10 billion in the last year. And that was our first 10 billion year. The year before that, we did five. And so we went five, 10, let's say we did 20. Who knows? So maybe the 100 was a little understated. And I think the reasoning part of it is, is everything that matters to us in the final analysis touches the physical world and we live in the physical world. And so I think the market opportunity for bringing software, computing, intelligence into an industry that largely historically was mechanical, electrical, that's a big opportunity. I mean, look at obviously what Elon has done to the to the automotive industry turned it into essentially a computing industry, a technology industry. And so I think every car company is going to be a technology company. Every shipping

and receiving company, logistics companies can be a technology company. I'm super excited for FedEx. They're obviously going to be an AI company. They're obviously going to be a technology company. Walmart, obviously going to be a technology company, an AI company. So super excited for them.

The entire telecommunications industry will be reinvented again into a technology industry. They were a technology industry a long time ago that became a services industry. They're going to be a technology industry again. And I'm delighted to work with AT&T and Verizon and Europe around the world as they reinvent themselves. And so you take a look at every one of these industries that touch the physical world. And there's a great idea for reinventing them with AI. I was just in Washington. It's so weird. You know, people put us in chips. Yes. You know, and, and so we're actually in warehouse and logistics. Yes. We're in healthcare. Yeah. We're in automotive. We're in telecommunications. Every industry on the planet. Yeah. Every industry on the planet. And, and so, so we're in light, we're, we're in, we're in life sciences, you know, and so, so all these entertainment, I mean, obviously for a long, for a long time and video was really.

vastly entertainment company. So each one of these verticals, NVIDIA has a massive stack that's inside of it and we're very deep in the ecosystem and we have excellent partnerships there. And so one of these days, people might not categorize us as a chip company anymore. The playbook would be to take the Alpameo automotive. I guess playbook and insert into each one of these industries. Yeah, that's the final contribution. But in each one of these, it takes three computers to build a physical AI platform. And so you need the computer to create the brain. That's the training system. But very importantly, and this is where you're going to see the labs also do this, they need to build a digital twin of the internet.

so that they don't have to put their AI agents out into the internet to learn how to use it. And so just like we created Omniverse as the digital twin of warehouses and buildings and cities and the physical world, we put our robot into the digital twin. The digital twin learns how to be a robot. And then we deploy it into the last computer, which is the computer that's inside, for example, a robot or a car or a surgical instrument or whatever it is. And

so it takes three robots, three computers. and three complete stacks. And we build complete stacks for each one of them. We're the only company in the world that does that. And then if you're a company that wants to build it end to end, you could use two or three or one of our computers, we don't care. And so each one of a computer has its own software stack. We do the same across all of our verticals. But of course, these capabilities cut both ways, which is why we hear a lot of people talking about safety. You've been highly critical of the doomer.

narrative. You can only get there if you build the technology responsibly. That's right. And so the first thing is, build the technology responsibly. Notice the word build. It's an engineering word. It's about technologies, about engineering. Engineers, scientists, researchers, we have to build this technology safely, responsibly. And the question is, can you? And the answer is absolutely yes. Because if you can't, then you shouldn't build it at all. Of course. And so we either have to believe we can or we shouldn't build it. OK. And so I believe we can. And so let me just reason about it with you. So the first part of it is they say some people say that that AI is uncontrollable because it's impossible to align perfectly. Alignment means this for all the audience. You tell somebody I want you to do this. And so the mission is clear. The mission is clear.

And it's clear to you because you said it in English. And so it could be, I want you to solve this quadratic equation. And so the mission is to solve the quadratic equation. The alignment is how do you do it? There are many ways to do it. For example, you memorized it. And so the answer is whatever it is. Or you saw the answer in a textbook, so you looked it up. You refer to a reference. Or you went to a drawer and you know that drawer had all the answers. You opened the drawer, you picked out the answer. Some people call that cheating. I call that opening the drawer and retrieving the answer. And that's what computers do. There's a lot of different ways you could solve that problem. Now, of course, that one is very explicit. Here's another one. Be successful in life. The mission is easy to say. And in fact, we say it to each other all the time. We say it to our children. We say it to our friends. Let's do our best to...

contributed society? Well, now the question is how? There are so many ways. How do you reasonably align in everybody's culture and background? And some people think that contributing to society means to be generous, to be kind. Some people might mean that contributed society make a lot of money and then be feeling tropic. Some people might so on and so forth. Isn't that right? Yes. Okay. And some people might just think that building a company alone is or doing something alone is a sufficient contribution to society, creating a beautiful piece of art or all of it is true. Correct. And so the question is, what's the alignment? And it just depends on your background, your culture, maybe has something to do with where you live, where you came from. There's just infinite number of ways. And so my point is this, alignment is going to be a forever problem. Yes.

If we depend on perfect alignment in order to ensure safety, then we're never going to get there. And so the answer is this. The way we treat, the way we think about babies and when we were raising kids and, you know, we have pets or whatever it is, employees, okay. You start with containment. You give them the rights that the access to tools, the access to information, the access to whatever responsibility, capabilities that they have, funds, resources, you give them access based on the job that you ask them to do. You start out with zero trust. When you hire somebody, you start out with zero trust. When you have a baby that's one year old, you put them in a playpen, you assume zero trust. And so you zero trust, you give them minimum rights and you expand that boundary as you learn more. And so whenever we have a new AI, we have to assume the same thing.

Whenever you're testing an AI, zero trust, minimum boundaries, minimum rights, and you expand it as you test and give yourself confidence. So containment, very tight containment is really important to any autonomous system. Second thing you do, monitoring. And the monitoring has to be out of band. And the reason for that is because if you're an intelligent person, hey, listen, I'm going to sit here and watch you do your work to make sure you're doing it right. And I'm going to sit here and watch you make sure that you're serving customers properly and that you're working hard enough. Okay. That and that autonomous system is going to behave differently than when you're not watching. It's like bras in the room with me or not. Exactly.

Exactly. Okay. Yeah. And so, so the, the monitoring has to be out of band and it has to have the ability.

to observe that the autonomous system is doing something that was inconsistent with the policy or inconsistent with alignment. And you have the ability to either escalate or shut it down. Right. Just turn it off. And so those kind of capabilities have to exist. That's engineering. That's technology. And it's possible. And we announced it recently called OpenShell. We have over 100 companies now joining us. We've got lots and lots of activity around the world. Because everybody wants the same thing. Starting at the highest level. You know, when I looked around the table at the White House, everybody around that table, right, their parents, they have kids, I think there's this view in the country that there are some leaders of superintelligence that are just yoloing, that don't really care, they're being reckless, they just want to grab all the money. And yet when I looked around that table, There wasn't a person in there that I saw that wants to be reckless, that wants to yolo. In fact, everybody there wanted to find the right balance between fast and safe. That was the president's directive. And we debated a little bit what that meant, right? There are different versions of how to implement that.

But what would you say to the American public who fears that we have a reckless president or reckless leaders? Did you see any evidence of that? You have kids. Do you worry about the future that you're bringing into existence with super intelligence? Safety and security technology are part of AI. There's some weird formulation that AI, technology, and safety and security are orthogonal. It's kind of like the airbags and the braking system and the radars, the cameras, the ABS, the lane keeping. All of that technology is somehow independent of the car. Do you guys understand what I'm saying? The only thing that matters to the car is the engine. In fact, the whole car is part of the car's technology. When I say we need to accelerate AI's technology, what I mean by that is we need to accelerate the security technology. We have to accelerate the safety technology. They all have to be accelerated. If you ask me right now, I think everybody would say, let's accelerate security. Let's accelerate safety technology. Nobody says, no, let's paste that. We should accelerate that. It's all part of the same thing. You can't separate capability, functionality, from safety and security,

they go us one. And when you deliver the service, nobody's gonna say, you know, I really like the functionality of your service, but boy is it, you know, wrecking havoc. So I think it's all one thing. And we treat it as all one thing, and the industry has to treat it as one thing. And when we say accelerate, we have to mean accelerating.

the entire package. And right now, including, right, especially the same. You mentioned pacing, especially, you know, this, some people read pacing the frontier by Dario and thought that meant slowing down. And I thought a funny moment in the room was when the president called on his new best friend, Dario, he referred to him in that way. And he said, you know, explain to me, you know, your version of fast and safe. And Dario started off by saying, Mr. President, you said whoever wins AI. wins, right? And he said that we're all speeding up, that they're speeding up, that they want to go fast, that they want to, you know, win the race with China, but also acknowledge that along the way, we just, to your point you just made, we have to make sure the safety systems are also speeding up and also keeping pace. So I thought that was an important moment where you had, you know, because I think people have positioned him as though he's an opposition right to this, but it seemed to me that you were all very much on the same page. You went to the cabinet room with him and Greg and the other frontier labs to sign this agreement. Was there any chatter in that room about differences of opinion or was everybody, you obviously got it signed. Did anybody have to be put in a headlock or did everybody go willingly? No, everybody was very constructive, very productive. And I think that...

Without exception, everybody is dedicated to advancing the industry safely and securely. And the reason for that is because if it's not safe, it's not secure. The functionality will never get a chance to speak. And so the capability will never get a chance to shine. And so we know that the priorities are at all times safety and security. Of course, recognizing that this industry came through a transition just recently. It wasn't but six months ago. this technology barely worked. And so the functionality, without functionality capability, what's the point of making it safe and secure? Nobody even cares about it. And so obviously the first job of AI is to demonstrate it could do something useful. And it really reached that point six months ago. And so now it really wants to be deployed and broadly

deployed. And so there's no question that the industry, the entire industry now has a strategic imperative, an incredible imperative, and great urgency to make sure that whatever we deploy is safe and secure. So I think everybody is completely convinced, aligned functionality, capability, safety, security, they're all at exactly the same place, priority number one. And so the question is winning. I would reframe winning.

not of the technology industry winning, not of just the AI industry winning. America winning means every industry in America winning. That's right. Nothing would give me more joy. The definition of America winning is when we talk about Lily winning, Wal-Mart winning, FedEx winning, every single one Boeing winning, the US military winning. Every citizen winning. Every citizen winning. our universities winning, our high schools winning, that's the definition of winning. The winning, there is no winning if it's just the technology, but nobody used it. The greatest risk is, of course we're gonna, I'll just take it, I'll just say baseline. There will be no technology industry if we don't build it safely in security. This is an existential issue for us.

And if we make it an existential industry for everybody else, well, that's not going to happen because it would be an existential issue for us first. So having said that, the definition of winning for us, and this is the part I always have to change a little bit. I want us to pivot. I like that. I want us to pivot. And when I see AI winning, America AI winning, I'm looking down from. you know these days we spend a lot of time in the air and we're looking down every state's winning every county's winning every community is winning every company's winning everybody won because we could afford to spend more on health care we could afford to spend more on education we could afford to spend more on interstate you know commerce highways etc all of the infrastructure of the country what I hear you saying is the definition of winning is that when we look back 25 30 40 years from now the economy needs to be 10 times as large everybody's lives need to be better and existential and primary to all of that. That's right. It has to be done in a way that's aligned with human interests, which this is the last question I have on this, and I know we have a lot to get to. Most people don't understand what our security framework is in the country. I've said we have the most robust super-intelligent security framework in the

world. We're clearly the only country on the planet that's actually sanctioned a model and pulled it off the shelf with mythos earlier this year. But I describe it as a three-legged stool.

We have the frontier model review of every new frontier model by Casey and the Department of Commerce. So they do all their work. If new models coming out, they review it, they test it for whether or not they think it's going to cause a problem in the marketplace. Now we have the White House accord on super intelligence that you help to architect internal and external audits, board oversight, and all of the liability that comes with that. And then third, we have the SI force. now led by Jay Clayton, of course, Emile and Scott and Andrew. And that's doing critical industry coordination, cyber, et cetera. So these three legs on the stool, there are people out there. I heard Gates recently saying, we're not doing enough. We need new laws on the books and everything else. That three-legged stool seems like a lot to me in a very short period of time that's come together as a leader of one of the most important companies, obviously, in superintelligence. Do you feel comfortable?

right, for that future that we're trying to usher in, that is a good step forward where we need to be right now. Yeah, the best way to get to the future is take action quickly. Yeah. And if we argue about what's perfect versus what's first, we'll end up arguing about perfection and we'll never get there. And so, I love the fact that these three-legged stool are actionable. They're tangible. They have real teeth. Happening now. It's happening right now. We're doing it now. We're taking action now. And the reason we're taking action now is because it's good for our company. It's good for our industry. It's good for our country. It's good for society. It's good all the way around. And if we don't do these things, The outcome is less desirable. We don't get to pursue that optimistic future that we all dreamt about. And so if you go back and look at another industry, I don't remember another industry that is currently at its current state and have now formulated all of these cross-company, cross-industrial government to industry coordination.

to make sure that we advance this in a responsible and optimistic way. I just think it's so important for people to understand because when I hear senators talking about it on television, on both sides, to be perfectly honest.

Well, they have to learn about what we're doing here first. Right. They don't even know what we're doing. Yeah, yeah. Right. Like we are doing an extraordinary amount. And by the way, these are voluntary things. There's not going to be a frontier model that gets released that Casey is not reviewing and the Commerce Department is not reviewing. And they demonstrated they have the authority under export controls to pull them off the shelf. Two things. One, come and spend some time and learn what actually has to be done. What's the problem?

Right. If you want a solution, you got to at least understand what the problem is. Right. Right. And so, so one, come and just learn about, we got plenty of time and not all the time in the world for this kind of stuff. And just, you know, come talk to us, come talk to our industry and learn. The second thing is learn about the solution that we put in place and then, and try to see if we can make that, if there are ideas to make that better. Of course, we're all completely open minded. Right. You know, and so. But be constructive. I thought that was part of the fourth point on the accord on superintelligence was this is not static. This is dynamic. This group is going to get together routinely. We're going to update the framework as required when the technology evolves. But I want to move forward now to compute. Of course, long running agents that work in teams of thousands or millions to solve complicated tasks, it all consumes massively more compute.

Dylan, our friend, as a semi-analysis has said, we're going to bring on board 43 gigawatts of new AI compute next year. Just to contextualize it, we'll share a chart with people online. That's more than all of the AI compute that exists in the world today. So we're going to bring that on board next year. The year after that, another 40 gigawatts of compute. You know where every single GPU is allocated. So as you sit here, does that sound right to you? Do you think we're going to bring on that much compute? And do you think that some of the things we're hearing out there about permitting issues or bottlenecks in the supply chain or local protests, do you think the number is going to be materially different than that based upon what you're seeing? I think the demand is definitely there and more.

The demand is definitely there and more. The challenge, of course, is that there are a lot of obstacles. When you move from digital to the physical,

there are a lot of obstacles. There are land permitting. There's power generation that you got to bring online. Of course, you need construction workers and outfitters and plumbers and all kinds of... Created a million new jobs just building these things. Created a million new jobs. And their skills are not fungible. They have a lot of skills. And particularly the ones that are building these 50 billion data centers, you're going to want some people with great skills. And there's a critical part of every single one of the build out of these things where you need some very specific skills. And so I think that, and then there's, of course, whatever moratoriums that are happening, whatever issues that are that we're seeing around the country. And so I think my guess is that land power shell, the build out resources will be one of the most important obstacles to reaching that. And so our company, as you know, now speaking specifically for us, there's several things that we're doing to mitigate that. And one, You know, because we're the only company that serves every CSP. We serve every NCP. We have great enterprise businesses through Dell and HP and Lenovo and others. And because we serve every single industry and work across the entire lifecycle of AI, everything from, you know, of course, data processing to training to post training to inference to agent processing.

and all of the tool use that we accelerate, our exposure to our go-to-market is the largest of all. Every contract manufacturer in the world is working on my behalf. Every go-to-market is connected to architecture. And so we have one particular state, one particular community, one particular data center has a hard time. Our demand is so much greater than our supply at the moment. We'll just adjust our supply chain accordingly. The other thing that we have, and I don't love this answer as much, but as you know, NVIDIA is also the infrastructure for the planet. And so whether it's in Europe or... early developments now in Africa, in Central Asia, in Southeast Asia, in Latin America, in Middle East, and so on and so forth. And so I don't love that United States can't keep up. But for our business specifically, we have such a global reach and there's build out demand all around the world. If we don't build it out here, I think we just all have to recognize that every country in the world would love to have it built out there. And so I think that our business specifically is probably the most resilient. And the reason for that is all the things that I mentioned. We cover the entire spectrum of

AI. We cover the entire spectrum of computing. We're in every single cloud, every single region.

So the resilience of our company is much greater, but I do think that LAN PowerShell is going to be a challenging part next year. And I guess what I hear from the street is there's this idea that maybe there's 50 gigawatts of announced chips that are underway or talked about, and maybe 35 to 40 gigawatts of LAN PowerShell that might come online. Fits into that land power and shell and on what timeline is a question up in the air. Yeah, but I guess like if we think of this through the entire and just Clark just interrupt you there and and I You know, of course as you know NVIDIA's architecture is the most productive, right? It generates the most tokens per watt and so if the world's if the world's amount of watt power energy reduces it shifts their choices to the most productive version. And the reason for that is that they still want to make their revenues. And so all of my CEO colleagues, they all have high expectations for revenues next year and people have, they need to deliver it, their shareholders counting on the delivery, all of their partners need them to deliver it. And so when the amount of power reduces, the quality of the compute you put into it has to be the most...

most extraordinary. We're the most productive computing resource you could put in the world today. We're the most rentable. We're fungible and we're durable. And so the forces around the world because of the shortage of resources is shifting interest towards us. Our market share was already very high. But surprisingly to everybody, not only are we a large company and our market share is high, our market share is actually increasing for these reasons. And so, so you're going to see us go from, and because we offer so much more per gigawatt than we used to, we used to be a GPU company, then we became the largest and fastest growing networking company. We're now going to be one of the largest CPU companies, surely the fastest growing one. And so, so because we're offering more and more, and we're into, you know, we're in computing for storage, we're computing for the knowledge infrastructure. And so the amount of, and now we're in security with open shell and sentry. And so, so the The the offering that we we offer into one gigawatt went from went from I guess about you know, 18 billion or so per gigawatt to 20 somewhat billion per gigawatt to now 40

plus per gigawatt And my sense is that that's going to keep on going up and so so not only are we are we growing share because because people are shifting towards our computing platform We're also growing share because we offer more to the world's capex and into the world's a dollar dollar share of the market. Yes. If you look at just go back in just six years ago, our percentage of the world's CapEx was in single digits. Then I think right now with the world's CSPs, we're probably running about 40%. But we're also expanding the market beyond the CSPs. And outside of the CSPs, whether it's on-prem or Neo Clouds, half of our business, our percentage of CAPX is probably what, 80%? Yeah. And so between half of our business being 40%, half of our business being 80%, and both of us growing and doubling every year. And so you could see that our percentage of the world's CAPX is in fact even growing. Yes. Well, one of those buyers of NVIDIA somebody we talked about two years ago, Elon, when he started building Colossus. And we were blown away a couple years ago to see him stand up a couple hundred thousand hoppers right in Memphis. And I remember you saying nobody can do it like him. Now he's raising another 40 billion going to buy maybe million plus Vera Rubens. He's moved from the case to the end. He says he wants to have 10 gigawatts by the end of 2027. That's hundreds of billions, right, spent on Vera Ruben. It's an unprecedented rampant scale. You were with him again yesterday when he was also presented with a medal. Is Elon that much better?

at building compute today as he was then, is it now obvious to you, is it a foregone conclusion that SpaceX will be the largest AI hyperscaler in the world? Well, we used to, in the two parts of our chart, Elon used to be under enterprise. That's the way we reported the numbers. Now he's part of our CSPs. Of course. Yeah, yeah. Of course. They're massive. So I'm delighted that... I love working with Elon. He's using our technology for his own AI in training models and such, deploying models, of course. Grockbot is super cool. And the demand for Grockbot, the consumption for Grockbot is really skyrocketing. And then, of course, he rents it. And so the combination of these three things, he's fairly unique in doing that. I guess Google does that. That would be yet another example.

And so I think he's also has the advantage that he loves the intersection between computing and the physical build-outs. He loves that whole area.

And nobody knows more about energy than he does. And one particular new market that we're going to go explore together is putting data centers out in space. Two engineering teams are working super closely together. to create a new type of server, a new type of architecture, that might be super good. you know, in space and on Earth. Well, we saw that Starship launched last week, you know, had its first orbital rendezvous and was launching those V3 sats that literally captured the imagination and kind of broke Twitter for a moment. Yeah. Just people mesmerized by watching these launch into space. I can't wait to be a payload. I mean, not a person. Do you think, does that give, do you have increasing confidence that we will have a material amount of compute in space you know, via Starship, let's call it 2028, 2029. I mean, Elon's put some big numbers out there. But do you think that we can get to material inference in space in that timeframe? Only NVIDIA compute. Only NVIDIA. Let me tell you why. It's not just about the technology. NVIDIA's performance per unit energy is the world's best. And so we call it performance watt.

And, but the most important thing is, is this, the productivity of NVIDIA's compute is so good, meaning the return on investment, invested capital was so good. Yes. That you could put it out in space and the launch cost is a small percentage of it. The launch cost is not any consequential. Right. And so you're not going to put low cost computing. Correct. Correct. You're going to put high value computing right there. And so where the highest value computing system you could imagine, the highest computing system in the history of computing. And so we're the perfect fit, really, to launch into space right now. Irrespective of the launch cost, it'll still be excellent R1. So do we get 500 megawatts in space by 29? Well, I'll let him tell you. But I think. Is it possible? Elon's launching 100 launches a year right now. I don't know what exactly it is. It's going to go to multiple a day. Yeah, right.

And so you could put Vera Rubins in there and it'll justify it. The demand is there. His return on investment is so good on it. And so the productivity is excellent. It's fungible, meaning you put a satellite on space. If you don't have a customer to well rent it, it's societal and you certainly don't want that. And so NVIDIA Compute is going to be fully fungible. And there'll be lots and lots of renters. We work with every single closed model company in the world. We run every single open model. in the world, we have

thousands of startups working with us. And so enterprises around in every single industry. And so the computing is completely fungible and is durable. You'll last, you'll sit up there and we'll keep up. And the reason for that is because we keep CUDA upgrading the software. We're the only hardware company in the world that you pay us eight years ago and we're still updating your software, new algorithms. And so we'll update the software for as long as the satellite's up there.

And so it's going to be really durable. And so I think the combination of these three traits along with the technology makes it the perfect payload for SpaceX. It costs trillions of dollars to build all this CapEx. You guys have leveraged your own balance sheet to help folks like the Neoclouds build that out. In order for finance people to get it to pencil out, they need to see payback, right? You talked about some companies seeing one-year payback, 50 billion in the first year, some spot pricing out of SpaceX, perhaps. But if I look at the total incremental offtake, the revenue of all the labs this year. So if I add up the two frontier labs, oh yeah, get yourself a snack. And we add up some of the inference labs. It's about 200 billion.

We need to see that double next year to 400 billion. We need to see it double the year after that to 800 billion, right? In order to get in the zip code of the payback on trillions of dollars of CapEx, we had this massive unlock this year with Claude Code that led to a 10x increase in Anthropics revenues. What's your confidence that the labs can continue to 2 to 3x their revenues every year in order that we get the payback that's going to be required just to build out trillions of dollars at CapEx. I mean, you're investing in these companies, you have big investments in these companies, right? Do you see that level of growth continuing in these companies? Oh, yeah. People misunderstand what's going on. So first of all, particularly, I mean, NVIDIA represents, call it 80% of the revenue share of AI infrastructure today. And so my description of the universe of computing and video computing matters here. They really basically call it three segments, three large segments. It's probably slightly more than that, but let me just kind of categorize in three large segments. The first segment of our computing is just baseline computing, just meaning used for data processing, molecular dynamics, physics, simulation, fusion, simulation,

fission, simulation, you know, you name it, right? Quantum chemistry, quantum physics, you know, and so everything.

Right. Okay. The world of simulation and data processing, not to mention computer graphics. And so all of that processing done on our GPUs, if you just kind of said there was no AI at all, just classical machine learning, XGBoost, that visa, my mastercard, you know, for fraud detection, recommender systems to figure out what news to recommend to you. It's all of that stuff combined. Classical computing, not AI, not deep learning, not agenting systems. I'm going to call it a third of our business. Yeah. And that business is growing, you know, call it 30%, 40% per year. Okay. In the absence of any AI labs, NVIDIA is not zero, right? We're quite a large company. And so I had number one. The second part is the AI labs, people see two, but we work with every one of them. And Metamuse is doing fantastic, right? And it's vastly NVIDIA. And, you know, There were zero last year, right? You know, a year before that, it was just recommender systems. And now they're, you know, killing it with metamuse. And of course, GROC didn't exist three years ago. We didn't run Gemini before we run Gemini now. And starting this year, we started working with Anthropic. And so if you just kind of look at our compute, people see a couple of labs, but I see more than that. I see five, six labs. And so these labs are all doing great. And if you told me that that represents not quite one third of our business, let's say a quarter of our compute, fantastic. So let's say that's a quarter. So when people are looking at 80 NVIDIA and we now run the infrastructure of the world, and I just said 25% of it, maybe 30% of it is the frontier labs. Call it a third is just processing in general. And another third, this is the part that's really great. Because we're at the beginning of a new computing platform shift, just a vast number of new startups are coming out. So many great companies. And some of them are, of course, doing consumer stuff.

you know, one of them really doing great. Noah instinct. I was just going to mention that. Noah's doing great to all kinds of physical AI, biological AI, material sciences AI, Jeff Bezos, this new company Prometheus, they're doing fantastic. So the number of companies across the spectrum of industries that have been started, thousands of them. And what do they do? They rent NVIDIA computers. So they're renters. And so what is a third of

our business is cloud computing for the next generation. It's kind of like AWS for today. And so it's just not AWS for the future. And that represents about a third of our business altogether. My sense is that we're in the early innings of the build out of this new computing architecture.

And it's a real big mistake to recognize just a couple of companies and think that they were funding the infrastructure. And then one observation, open models. You know that I've been a great advocate of open models. And the reason for that is because open models opens up new opportunities. And if you look at all these startup companies, 80% of them use open models. And they're not using it to compete with the frontier labs. They're using it to unlock a new opportunity somewhere else. You know, it could be in physical AI, it could be manufacturing, it could be healthcare. They're doing something completely, completely different. And they're taking these open models and many of them taking Nemotron and these models are then customized into super intelligence. And so if you will, I kind of see the...

the general closed labs will be artificial general intelligence. They'll get super good. And then you have all these vertical companies that are focused on a particular domain and they have very proprietary data, they teach you to do something very specific and they achieve super intelligence for their domain. So you don't see, you know, so again, If we go to your compute math, which I think a lot of people are using to forecast the future revenues of the labs, for example, you said, hey, if you stand up a gigawatt of this compute, you can get 40, 50 billion out the other end. But if we took the number we talked about earlier, 40 gigawatts of compute coming online next year, and if you took 40 billion out the other end, that would be 1.6 trillion of revenues coming out the other end of that factory.

But of course, it doesn't work that way. At the end of the day, you've got to look at the demand curve and the rate of diffusion to determine the revenue that's going to come out the other end of the factory. And what I hear a lot is that people are nervous. They've seen over the course of the year maybe the lab's revenue is slowing down a little bit. And they're nervous that at some point you're going to have the proverbial bubble popping, where we have more supply coming online than we have demanded a moment in time, that AI is more like these step functions rather than just this parabolic line

straight up. I guess the question to you is, are you worried about a moment in time where everybody gets nervous about financing the build out, and then we don't get the build out? Well, all these different things, you just got to break it down. Break it down back down to first principles.

The part that people are trying to extrapolate, the 100 billion of AR or 125 billion of AR they're trying to extrapolate is a small part, not a small part, it's just the part of the overall infrastructure build out. And so the question is, are people going to continue doing shopping online? So that third of the build out, let's call that a third. You said it's third of your business. Take that to the bank. Yeah, right? That's not even a debate. So if you say, you know, I said a few, was it a couple of years ago that as we exit this decade, infrastructure build out will probably be somewhere between three to four trillion a year. You know, obviously it's closer to four than it is closer to three. And per year, by the end of the decade. Yeah. And if you said that one and a half trillion of that was just compute, just compute.

Now, why is it going to be so big? Well, it was half a trillion before I started. And if I told you that over the course of some six, seven, eight years, it became one and a half trillion because the world's just consuming more compute. I mean, it's not illogical. On top of this, remember, Moore's Law is over. There was a time, if you want to get twice as many transistors, you just wait one year. And if you want twice as many transistors, get another wait one year, same price. But if you want twice as many transistors now, it doesn't matter how long you wait, you have to pay twice as much, okay? Because Moore's law is over. And so whatever computing capacity that the world wants, every single year, if you want twice as much, you're gonna pay twice as much. And so on top of that half a trillion dollars that was kind of growing just naturally, because the world just can use more and more digital, you're gonna have to effectively, you know, give it an X factor.

just to account for the fact that Moore's law is over. And so without AI, the semiconductor industry was on its path to become the largest industry in the world. And I've mentioned that long ago, just people forgot, which is one of those things that I said. I just said that two years ago, yeah. And so that's just one part. And so the only thing that we, and then don't forget, because we're in a new computing platform shift, there are gonna be a whole lot of

startups. Innovation is part of the economy. New companies. And we said it's accelerating. Yeah. So if we spent, if you go back to Moore's law, general purpose compute, 5 billion per gigawatt. Today it's 50 for a Vera Rubin. So the question is, is Vera Rubin deliver 10x more value in the world? The punchline is, I don't think it's necessary to extrapolate any of the labs into multiple drones of dollars. Unnecessary to do that. I think that's a, I mean, it's an important one because I think people You know, we just saw yesterday, you know, even in the market, you know, some chatter about was OpenAI's revenue 70 billion or 50 billion. I think you and I, they're going to be insanely successful companies. Correct. Yeah. But you saw the pullback in a lot of the infrastructure companies, right? On the fear that, oh my gosh, OpenAI doesn't have the money, won't have the money in order to pay for the infrastructure. I think the point that you're hammering home.

is that when we look at the 40 gigawatts of infrastructure built out next year, at least a third of it is just going to write general-purpose compute replacement, the basic stuff out of the hyperscalers, that a third of it's going to kind of neocloud, sovereigns, enterprises, everybody else who's using this, all the startups, and a third of it's going to not two labs, but to many labs, including GROC and thinking machines and all the other. We're putting way too much pressure on two labs. Yeah. I'm delighted to partner with them and support them. They're going to be two of the most consequential technology companies ever built in history. They're going to be magnificently successful. But they don't have to be anywhere near a trillion dollars in revenues to be successful. I think we're putting way too much pressure on that. And then the rest of the industry, we still do have to build out a few trillion dollars of computing infrastructure. Just to keep going.

Listen, let's shift a little bit. How good is this, guys? Come on. By the way, what are these called? These are dried plums. Dried plums, the old dried plums. Yeah, this one particular store in Taiwan is one of my favorites. Kind of a licorice-y. There is licorice, yeah, licorice, a little licorice, salt, sugar, cinnamon. Yeah, that's a little twang to it. Yeah, yeah, you got it. Yeah. Okay, you set us to next year demand for NVIDIA's chips well over 100%. You guided to 70% revenue growth. You gave a year forward look because you said supplies constrained. That's all I had at the time. That's right. Yeah,

I got 18 months to work. Right. You said you were very confident. I'll come to work every day. You said very confident and plenty of time to work with the supply chain to do even better. You expected to do even better and not disappoint customers.

So are things going even better? And has the order book continue to grow? Because the other part of that is, right, you took a snapshot at a moment in time. And of course, people are continuing to place orders. So do you think there's enough slack in the supply chain you can squeeze more out? And is the order book continuing to grow as more of these capabilities are unlocked? For the moment, I'll tell you why I told everybody. The challenge that we have is we have giant order books. Every single CSP every single NCP every single OEM had already placed orders on us startups big company small company They already place orders on us And the reason for that is because we work with them on it on it depending on on the size of the customer at least a year For the the largest ones two years forecast and so so we already have a very large order book and because I was in the process of allocating and What I was allocating to all of our customers is a lot less than... They want it. They want it. If I didn't publicly say what we had, everybody would be second guessing, are they shortchanged, are they being... And so it was my way of essentially handling every single conversation where I'm disappointing people.

And so pretty much every customer would buy more from you if you could deliver more. By far. And then since then, of course, we have new orders almost weekly and daily. And so in addition, however, we're getting more supply. And some of the supply that is limiting us is land power shutdown stream. And so a couple of weeks ago, we announced two gigawatts in Australia. And so that's a very big deal. Two gigawatts to us is like 80 billion. Huge. And then I'm hoping when I go to Europe with GTC, we'll be able to announce some new partnerships there and unlock more land power shell. And so this land power shell is clearly one of the obstacles. And this is an area where NVIDIA has a huge advantage. Because based on what we build, if there's a Neocloud and this Neocloud-like core weaver and EBS, And they would like to scale out their business. NVIDIA has the entire full stack, complete infrastructure. And so we could support them as their scale out. And so we could build in places and we can build with partners and

customers that nobody else can. And so I think this gives us a lot more resilience, but nonetheless, as we continue to increase supply upstream and downstream, One of the critical areas is substrate. Substrates are getting larger and they're more chips. And so it's working against them in two ways. Every single chip that's being shipped uses more of the material. And so these organic materials are really hard to make up the precision that we need because as the packages get larger, the thermal coefficients And the way we stack things make it really hard to handle things like Warpage and various manufacturing defects. And so at the scale that we do, we have to be very precise and we need really great partners. And so that's supply-constrained. And those are just some examples. Silicon Photonics is supply-constrained. So everything is supply-constrained. And so we're going to keep working on it. I have 18 months to do so. Every single week that goes by, we increase.

But nonetheless, every we keep increasing more supply and order books keeps going out. Right. And so it's going to be hard to close that gap for a few years. Take a look at this chart. You know, this is like a deposition. No, it's OK. Excuse me. Now, usually it goes like this. Mr. Cuomo, please. Please now. Please now look at the deposition. OK. So the gray line, this is was the consensus at the start of 2025. This was the consensus for Nvidia at the start of 2025. The blue line was altimeters forecast way back then. So we were a lot higher than the consensus. The green line is now where the new consensus is based upon your guidance. And so one of the things that blows my mind about this is the 25 people who get paid to just forecast what this company is going to do. Their consensus of all their ideas, right, was off by a couple hundred percent, right, just at the beginning of 20.

No, at this time last year, not even 25. That was the consensus at this time last year. Okay, one year ago. So in a year, the consensus forecast has shifted up by 100% to 200% for this company. Extraordinary. Like in my 20 plus years of doing this, you don't often see consensus being off by that much. And yet the stock's only up 20%. Okay, you literally doubled what the revenue forecast was and the free cash flow forecast was for the company. You're now trading at 13 to 14 times earnings depending upon who you're looking at. So what do you think the market is saying about Nvidia or about the confidence in the expectations of the market, right, to

cause that to be the case? Because usually if you see it shift up by that much, the stock would have shifted up by a lot more.

For a company to be as large as us gaining share tells you something about the resilience. Yes, and The the size and scale of our advantage, right? And and is that advantage increasing or decreasing is your mode increasing or decreasing increasing increasing you're a competitive mode Yeah, is increasing at scale. Yeah, let me give you an example Let's just do a thought experiment right now. Yeah, if you had to hand somebody a hundred billion dollar checked them place a purses order right now For a next generation. Yeah If you want to get in line for Vera Rubin Ultra, it's close to placing order time. Side unseen, right? But they have the confidence. And so let's say you got to place 100 billion on somebody. One, how do you know it's going to be any good? But because NVIDIA builds it, you know it's going to be better than Vera Rubin. And Vera Rubin's awesome. And the reason why Vera Rubin, the reason why everybody had confidence in Vera Rubin is because Black Hole was awesome.

And the world of computing, the workflows that these computers do are so complex. You can't just have like measure one benchmark. Right, right. You know, it's 50 billion. Yes. And the number of technologies that have to come together to deliver the performance is insane. You know? Yeah. And so the question is, sight unseen, who can you place a 50 billion order on? Yeah. We're literally the only one. Yeah. Now back when it was place appeal a billion dollar PO that was challenging enough, but you've had place a hundred million dollar PO and The the system is sitting right in front of you. You could just bench you could benchmark the chips Right, right. Yeah, how do you benchmark these systems? Just building it is a miracle, right? You know, so if somebody wants to do an evaluation I'm happy to support it But you're gonna need to give me a giant data center and we're gonna need a couple years to retrofit it and you know And so you can't even benchmark these things anymore. Interesting. And so it's exactly what happened with TSMC. The reason why we give 100% of our business to TSMC is because they're the only company in the world that I can plan five years out. Right. And know it's going to be. Insight unseen know that that transistor is going to be really good because it's better than that one, better than that one. Yeah. And that they will have capacity for me and that they

know how to work with my business team, that they have the agility to handle all of my ups and downs and, you know, for example, to deal with the fact that I went from grade to blue to green. Exactly. Okay. Exactly. They have to be big enough to help me go from grade to blue to green. Exactly. There's no other company in the world that can help me do that. That's right. And to have all these different process technologies, from silicon photonics to logic to memory to push, pull, push, pull, packaging. I mean, the number of things that we do is so dynamic.

And so how do you possibly do that with a company that is one-tenth their size? It's impossible. And so I think the big idea is this. Because we're building infrastructure for the planet and NVIDIA has become an infrastructure company. And because these infrastructures cost so much money and they're moving and technology is moving so fast, we're practically the combination of the trust, the fact that we run everything. You know, so it started out with some some GPT models But look at the flash attention mechanisms that were were re were invented the way that we paralyzed the models and sharded across MV link 72 and how all that changed MOE changed and now people are talking about Transformer diffusion models that are hybrid SSMs that are Transformers hybrids and all these different technologies are all right the evolution of all that We know that it all runs on CUDA, right?

We don't know whether you need to do more pre-training, post-training or just agent to inference. Well, it turns out we run it all anyways. And so you could build a 50 billion data center and know that the fungibility of the work and if something were to happen, you know, for whatever reason unknown, you want it to, if you will, sublease half of your capacity to somebody and video is completely fungible. So we're fungible in workload. We're fungible in models. We're fungible across customers. And we're deployable everywhere in the world. You start a company in Europe, you start a company in the United States, but guess what? You would like your customer, you want to serve customers in Europe, you want to serve customers in Japan. Well, it turns out we're already there. NVIDIA's infrastructure is already there. So, OpenEI wants to go to Europe, we'll take them there in the heartbeat. If they want to go to India in a heartbeat, isn't

that right? Anthropic wants to go to Japan, Korea in a heartbeat. We're already there. All of these architecture is already there.

If you take NVIDIA in the hole and then you come back and you say, okay, I want to do the thought experiment. I want to place 100 billion on somebody. How do you do it? A year ago, we were sitting here and the consensus forecast was that you are going to be flat or shrink in 2027. Now the consensus forecast is you're going to grow 74%. So I just want to let people let that sit in, because a lot of people follow these consensus forecasts, right? And it was really wrong, okay? They were forecasting. They go back and listen to your podcast from last time. Right, you told them. I explained it. Listen, ladies and gentlemen, I did explain everything. And so I look out at 2029. There wasn't an ROI question. Yes. And then there was this custom versus general purpose.

The reason why NVIDIA is general purpose is so I can have all this fungibility. The benefit of fungibility is low risk. And low risk really comes in to play when you're placing big dollars. When you're building out 50, 100 billion worth of infrastructure, you need to know that that infrastructure is going to be really good for a long time, no matter what. So specialization is the enemy of fungibility. Specialization is the enemy of durability. And so, so long as I can prove that our performance is extraordinary, and we prove that every single day, the moment we can prove that, then fungibility and durability translates to rentability, which translates to productivity. And finance ability. And therefore, finance ability, which leads to the lowest TCO. Right, exactly. Well, now you're buying back the largest amount in history.

And, you know, on my math, you know, you could probably buy upwards of 10% of the company back within the next few years. At Goldman, you called NVIDIA, you know, the first growth value company. Yeah. Sounds a lot like Apple's pitch to- Oh, is that right? Okay, is that right? Yeah, or more, yeah. Okay. That they could return the capital. Well, it worked. It worked, yeah. For sure. It worked. It went from 300 at the time, 30 to 300. Yeah. 300 plus. And in a lot of ways, we're going through a similar transition. For example, the thing I forgot to mention is, and Brad, I think you were about to ask me about that. Most of our investment and financing

partnership strategies is trying to set up two business models, two new business models. One business model we've been working on for some time and will be accelerated.

in the near future is software licensing. You know, the body of software that we have in our companies is extraordinary. And that data center is so large, imagine all the software that runs in it. And we provide the entire stack. Well, and our entire stack is completely industrial strength. For the CSPs, they want to run their own stack. And so we license it where we give them our software open source, and then they take it and they modify it and they integrate it into theirs, and then they operate. their services. But for a lot of the world's companies, they don't have the IT department, they don't have the software department do it, so they need me to maintain it. And so because AI in the last several years was largely focused on training, but now they're moving into deployment, they need to operate these as factories, and they need to know that I'm on the hook, I'm on the SLA.

to support it and maintain it, keep it secure, keep it safe for as long as we shall live. And so software license for our enterprise business will be very, very large. And so, you know, if our software licenses, you know, just roughly say for, for, for, for simple math, a couple of thousand dollars per GPU per year, and we have millions of GPUs, we have, we ship, you know, 20 million GPU say a year, I'm growing. And so half of that goes into a world's enterprise, it adds up. Yeah. Okay. And so I'm fairly certain this is going to be a very big part of our business software license. And then of course there are some clouds that rely on NVIDIA for our full stack. They rely on us for our offtake, our ecosystem. They also rely on us for maybe as an investment partner and helping them secure their financing and capital.

They've secured a lamp power shell their regional champion They got all kinds of stuff working in there in their advantage But these three areas they need our support on and so in those kind of Relationships is very likely that we'll move to ref share and so we'll make an investment We'll get the benefit at the sale up front But we'll also have the benefit of ref share going forward and so you know, we're not a cloud service provider But we're gonna be providing off-take and providing software support for them for for you know four, five, 10 years. And so we ought to get some economic

return on that. And so ref share is going to be a store sale. And so notice, I just described two new growth factors, independent of the hardware sale of our company. And it's much more related to the install base of our platform. What's the benefit of software licensing if you have no install base? And how could you have ref share if there's no offtake? And so the install base element, the richness of our software stack, the offtake, the richness of our offtake and the ecosystem that we have around the world, that's going to fuel that. One of the ways, you know, so we're talking about this trading at 13 times, you know, when Tim and Luca went to see Warren Buffett, and one of the reasons that Buffett has said he doesn't like technology companies is they have this hit.

They produce an enormous amount of free cash flow, and rather than ever giving any of it back to the shareholders, they go and spend it on something else. And oftentimes, the something else isn't as good as the thing that caused them to generate the cash flow in the first place. You guys are now spending over 50% of your free cash flow buying back your stock, effectively returning it to shareholders while you have this exponential growth at the same time. As we've already discussed. We said 50. Right. I think in this last quarter we did 60-some on. Correct. We invest in really high quality companies, as you guys know. The last couple of years, Whoever NVIDIA buys from, whoever Jensen buys from, you have to invest in that company. And now the new word on the street is whoever Jensen invests in, you have to invest in that company. And it's true. And the reason for that is because like you, we're in the ecosystem. We're in the flow of information. We do a lot of those investments together. Yeah. And we know a lot of smart people. And we want to back these people when they want to do something great.

And so we're kind of on the inside. We have a lot of information. And so our investments are well-informed. And not to mention, NVIDIA is also in the ecosystem. And so we can help them succeed. And so there's a lot of different reasons why the investments turn out great. But as a percentage of our total cash generation, it's going to decline very rapidly. Partly because OpenAI Anthropagin isn't going to come around again. That opportunity is not going to come around again. frankly, and then secondarily, our cash flow generation is growing quite fast. Right. And to that point, it seems to

me that if you're buying back that much, and the durability of the revenues, I think, as you've described here today, is longer than people expect. And I think that's the one place people are still hanging too much on this is only going to last as long as rest you this on the pod the other day how long until the bubble burst and I think you said something about two to three years you have good visibility that you know the demand is there and who knows kind of after that and I heard some people saying oh it may only be two years before this burst but when I look at this like this is fundamentally a rearchitecture of all global compute forever forever so unless there's a global depression yeah right then every single year People are going to have to consume an enormous amount of NVIDIA computer. Yeah. And so the way to think about that is we were running in the, you know, the CSPs were running about half a trillion. The whole industry, the whole world is running, call it a trillion dollars of IT build out each year. And every year more gets built and every five years or so, some gets, you know, depreciated or turned off and taken offline and so on and so forth. And some of them, some of it lasts longer. And so, so that, that old world was in the retrieval way of doing computation and the new way of doing computation. You know, I've predicted that going out of the, out of the decades, so three, four, let's say four, let's say five. And I, and my sense is that we're going to continue to grow from there. If you told me that the annual, the annual capacity building, infrastructure building, refreshing combination of new and old, refreshes, runs around 10 trillion. I'm not even a little bit surprised. Right. Right. Just again, for people listening, you think 4 trillion of annual spend by the end of the decade because every time you consume tokens, every time you use intelligence, you have to run the machine. The factory has to run.

I think you've described that. That's about 50%, if you assume that... 50 trillion of global GDP is related to knowledge work. If we get a 20% productivity gain over five years on that, like you've seen inside NVIDIA, you've seen much bigger productivity gains inside NVIDIA. But if you see that sort of productivity gain, then if we get 10 trillion of incremental economic activity by 2029, 2030, that seems very reasonable. And you're just saying, we'll spend half of that. on CapEx, on the intelligence, you know, to produce it. And so maybe, you know, that's a good place for us to take a bit of an off ramp. You've been incredibly generous. I could go on

forever, but I kind of want to wrap where we started, right? When we look back. If this was one of those eating, drinking podcasts. We could keep going. We could keep going.

It wasn't so long ago that the human condition was described as nasty, brutish, and short, right? Was that last week? Innovation has really driven all human progress, mostly in the last 200 years. More time with family, longer lives, less time doing drudgery that humans really didn't want to do. We're the first generation where 63-years-olds consider young. Exactly. Exactly. People call me young man. Safer, better lives. All of this despite all of these concerns that you and I hear out there. And by the way, those same concerns were expressed at the beginning of the Industrial Revolution. What's your level of confidence that in five years from now, we will have made the right decisions and the world will be that much better place that we describe because of the investment and all that we're doing. Are you more confident as we sit here today than you were 12 months ago? I'm 99.99% just in service in honor of a good friend not using zero and 100. I'm 99.99% confident that too many of us care due deeply that the outcome turns out well. That we will dedicate our might, our intellect, our resources to ensure we get there responsibly and optimistically. And so there's just so many people doing the right things. Now oftentimes, you know, there are people who pontificate about these things and they're informed by several things and they And they don't realize that there are a lot of people, industries, incredible researchers and engineers outside of their sphere of information that are doing amazing things all by themselves, independent of them. Just because they didn't say it doesn't mean a lot of smart people are doing smart things. And you know, I say this all the time inside the company. Even without my leadership, without strategies, the company is doing smart things all by itself.

Smart people do smart things. They do more impactful things when you organize a line, right? But Amazing people are doing smart things all by themselves. And so when somebody tells me that that you know, you're worried about it's good that we worry, right? And we should express our worries. We should express our concerns But we shouldn't predict the end of the world. Yeah, yeah, because they're underestimating the human condition They're underestimating the human capacity and they think for

some some reason with maybe it's just hubris, maybe it's arrogance, that unless they do it, it won't get done. Unless they do it, it won't get done. The fact that matters, this company is doing amazing things, whether I say it or not. And I get all the accolades and to blame, but in both cases, I'm fair, especially the accolades. And the reason for that is just these you know, when you surround yourself with amazing people and you create the conditions by which everybody wants the same outcome, the good outcome, the good version of it. And I think most people do. That's the reason why I say we need to hold ourselves accountable and responsible. But we should express a future that's optimistic. And I am 99.99% certain that surrounded by all the amazing people in the world, we're going to get there together. Yeah, agreed.

It's, I think, you're a national treasure. We're lucky to have you helping to lead the charge. The nation's indebted to you for getting everybody together, getting us all on the same page, building the framework. There's going to be more work to do. But as a father of two, and I know you have kids, you're going to have grandkids, and we don't care about anything more than actually building a better world for them, right? We could easily stop. But if we stop today, we lose. We lose the opportunity at that better world. So we'll wrap it there. Thank you, Jensen. Thank you, guys. And Brad, thank you for... I've never met a greater patriot, somebody who cares so much about the industry, but even more so the country. And your gift and your enthusiasm and just tireless energy in coordinating, pulling together.

a lot of people in order to do something great like what you did. Incredible. As a reminder to everybody, just our opinions not investment advice.